WP1 Connectome → SNN

M1–36 · PhD 1, PD1, PI · → H1

T1.1 electrical synapses, ♀ and ♂
T1.2 Tier 1 biophysical (fitted)
T1.3 Tier 2 reduced (KAN equations)
T1.4 Tier 3 trainable SNN
T1.5 motor readout, efference copy

D1.1 (M18) synapse atlas, ♀ + ♂
D1.2 (M24) reduced model + toolkit
MS1 (M12): Tier-2 error < 10 %?
MS4 (M36): H1 decided

WP2 Eristalis dimorphism

M6–48 · PD2, PhD 3, PI · → H2, H2b

T2.1 ♂/♀ model families (homology)
T2.2 stimuli that split the models
T2.3 tethered physiology and fills
T2.4 free-flight pursuit, high risk
T2.5 origin of the MIO: evo-devo

D2.1 (M24) discriminating stimuli
MS2 (M24): stimuli split families?
MS4 (M36): data done; go/no-go T2.4
MS5 (M48): H2, H2b decided

WP3 Theory

M12–54 · PD1, PI · → H3

T3.1 embodied simulator
T3.2 pursuit vs stabilisation, same wiring prior and energy budget
T3.3 ablation, mean field, KAN

D3.2 (M48) minimal motif catalogue
MS5 (M48): H3 decided; motifs → WP4

WP4 Active event vision

M13–60 · PhD 2, RE, PD1, PI · → H4

T4.1 active event-camera head
T4.2 motif SNNs on neuromorphic HW
T4.3 benchmarks: B-I small targets, B-II ego-motion, B-III quadrotor
T4.4 machine controversial stimuli

D4.3 (M54) benchmark + HoverEye data
MS3 (M30): head and quadrotor flying
MS6 (M60): H4 decided; releases

labels

statistics

models

WP5 Embodied validation at scale, teaching, open science

M1–60 · PI, student assistants, IU courses

T5.1 course modules every semester → benchmark entries and robot data
T5.2 robot zoo on low-cost camera robots → embodied robustness data (H3, H4)
T5.3 community proofreading of electrical synapses → T1.1
T5.4 hoverfly watch: pursuit statistics → T3.2 · T5.5 releases, evaluation